MINI PROJECT DOCUMENTATION

Smart Campus Problem Reporter

A Web-Based Campus Issue Reporting and Tracking System

Project Title	Smart Campus Problem Reporter
Development Tool	Base44 AI platform
Project Category	Web Application / Mini Project
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Project URL: https://radiant-campus-fix-flow.base44.app

Prepared for academic project submission

1. Abstract

Smart Campus Problem Reporter is a web-based solution intended to make campus maintenance reporting more organized and transparent. In a college environment, problems such as faulty lights, damaged classroom equipment, water leakage, cleanliness concerns, and Wi-Fi issues may be communicated informally. Informal reporting can make it difficult to maintain records and follow complaint progress.

The application is designed to provide a central place where students can submit an issue with relevant details and, where supported, a photo. An authorized administrator can review reports and update their status. Students can then check the progress of their submissions. The prototype is being developed with Base44 AI, a platform that helps create web applications from natural-language instructions.

This document describes the problem, objectives, proposed modules, system design, workflow, development approach, testing plan, expected benefits, limitations, and future enhancements. Verify the exact features available in the live application before final academic submission.

2. Introduction

A campus contains classrooms, laboratories, corridors, washrooms, electrical facilities, and network infrastructure that require regular maintenance. Without a consistent reporting process, information can become scattered across verbal conversations, messages, or paper records.

A web-based reporting system can organize reports in one place, help authorized staff prioritize work, and provide students with visibility into complaint status. This lightweight mini project demonstrates practical web application development and a real-world campus use case.

3. Problem Statement

Students may face difficulty reporting campus problems through a consistent process. Informal communication can lead to missing details, repeated complaints, limited visibility of progress, and difficulty identifying unresolved issues. A centralized system is proposed to collect campus complaints, maintain records, and support status tracking by students and authorized administrators.

4. Objectives

- Provide a simple interface for submitting campus problems.
- Collect issue title, category, location, description, and optional supporting photo.
- Assign a unique reference ID to each complaint, if implemented.
- Allow students to view the status of their own complaints.
- Provide an administrator view to review reports and update progress, subject to configured permissions.
- Present complaint summaries to help staff understand current workload.
- Demonstrate AI-assisted development of a practical web application.

5. Scope of the Project

The initial scope is campus issue reporting and tracking. Typical categories include electrical issues, plumbing, classroom equipment, cleanliness, Internet/Wi-Fi, safety-related facilities, and other maintenance concerns.

The application is a prototype for demonstration and learning. It does not automatically repair issues or replace official college maintenance processes. Real deployment would require institutional approval, defined administrator responsibilities, data protection practices, and user support.

6. Proposed System

The proposed system includes a student reporting interface and an administrator management interface. A student submits a report with the details needed to understand the issue. The application stores the report in its configured data store. An authorized administrator reviews it, updates its status, and may add remarks. The student can revisit the report to check progress.

Step	Action	Outcome
1	Student opens the website	Reporting interface is displayed
2	Student submits issue details	Form is validated
3	Application records the complaint	Complaint can be referenced
4	Administrator reviews the report	Issue is assessed and prioritized
5	Administrator updates status	Progress becomes visible to the student

7. Main Modules

7.1 Student Module

- Complaint submission: title, category, location, description, priority, and optional photo where supported.
- My complaints: view reports submitted by the signed-in student, if authentication is configured.
- Status tracking: see submitted, in-progress, or resolved status if these states are implemented.

7.2 Administrator Module

- Complaint overview: review submitted reports.
- Search and filtering: locate reports by ID, title, category, priority, or status where supported.
- Status management: update progress and provide notes where supported.

7.3 Dashboard and Summary Module

- Show summary counts for total, pending, in-progress, and resolved reports if supported.
- Optional charts can illustrate complaints by category.

8. Tools and Technologies

Base44 AI: AI-assisted platform used to generate and configure the application through natural-language prompts.

Web interface: Generated using Base44-supported capabilities. Record the underlying frontend framework only if confirmed.

Data storage and authentication: Use database and authentication features supported and configured in Base44. Test them before claiming persistence and role-based security are fully operational.

Web browser: A modern browser on a laptop or mobile device.

Live project URL: radiant-campus-fix-flow.base44.app

9. System Requirements

Hardware: Laptop, desktop computer, or smartphone with a stable Internet connection.

Software: A modern web browser, access to Base44 for editing, and Internet access.

Users: Students who submit reports and authorized administrators who review and manage reports.

10. Testing Plan

Perform the following tests on the deployed website and record the actual result. Do not mark a test as passed until verified.

Test	Test action	Expected result	Actual result
1	Open the website	Main page loads without an error	To be tested
2	Submit a valid complaint	Complaint is accepted and confirmation shown	To be tested
3	Leave required fields empty	Validation explains missing fields	To be tested
4	Refresh after submission	Saved complaint remains if persistence is configured	To be tested
5	Update status as admin	Status update is saved and displayed	To be tested
6	Check student/admin permissions	Each role accesses only authorized records/actions	To be tested
7	Open on mobile screen	Forms and navigation remain usable	To be tested

11. Expected Benefits

- A consistent and organized way to submit campus issues.
- Improved visibility of complaint progress for students.
- A central record to help administrators review unresolved issues.
- Potential reduction in lost or incomplete reports.
- Identification of frequently reported categories to support maintenance planning.

12. Limitations

- The application depends on Internet access and hosted service availability.
- Report quality depends on accurate descriptions and locations.
- Status tracking is useful only when authorized staff update reports consistently.
- Photo upload, authentication, database persistence, and administrator permissions depend on the live configuration.
- The prototype does not perform physical repairs or guarantee resolution times.

13. Future Enhancements

- QR codes in classrooms and laboratories to pre-fill campus location.
- Email or in-app notifications when supported and configured.
- Maintenance assignment to route reports to the relevant department.
- Trend charts for recurring issues and high-report locations.
- Accessibility improvements and multilingual interface support.
- Optional campus sensor integration, subject to institutional approval.

14. Conclusion

Smart Campus Problem Reporter proposes a practical digital approach to organizing campus maintenance complaints. Bringing reporting, review, and status tracking into one web application aims to improve transparency and make issue management easier for students and administrators. Building the prototype with Base44 AI demonstrates how natural-language development tools can support rapid web application creation.

Before final submission, test the live website, record verified features and results, add screenshots of the actual application, and ensure all claims match the implemented system.

15. Project Details

Live website: https://radiant-campus-fix-flow.base44.app

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